



MALAYSIA DATA INNOVATION TALENT x DOSM DATATHON 2025

AI/ML-Driven Rollout Plan for Data Centre Micro-Credentials in Malaysia

by

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1.0 INTRODUCTION

1.1 Background

The rapid global expansion of Artificial Intelligence (AI), Machine Learning (ML), and Big Data is driving the need for high-performance digital infrastructure. Data centres (DCs) are the backbone of industrial digital transformation, as they supply the computing power and storage capacity crucial for its success. Countries with advanced data centre ecosystems are strategically positioned to become leaders in innovation, competitiveness and growth of the digital economy.

Malaysia, which is currently a newcomer in the data centre market with approximately three years of exponential growth, has set future aspirations to become a regional hub. The Malaysian government has launched various initiatives to support this goal, one of them being the “Malaysia Digital Economy Blueprint,” which identifies digital infrastructure and talent development as key priorities. However, the sustainability and competitiveness of this industry depend not only on infrastructure but also on the availability of a competent workforce.

Micro-credential programs, which provide modular and industry-specific certifications, are an effective way to close the skills gap. Micro-credentials align the training supply with industry demand by enabling targeted and timely upskilling and reskilling. When executed strategically, these programs can significantly enhance Malaysia’s capacity to supply the DC sector’s talent pipeline.





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1.2 Problem Statement

Despite Malaysia's aspirations to establish itself as a leading regional DC hub, a significant gap exists in specialised skills related to DC operations and management. The current workforce development programs are not adequately targeting specific geographical areas and skill tracks for training. This mismatch may slow down Malaysia's drive to become a DC hub.

Without targeted interventions, Malaysia will lose its competitive advantage in the global market and investment opportunities. Furthermore, Malaysian employees will not have the necessary skills to work in facilities operations, networking, security, compliance and energy management in DCs. It is imperative to establish measures to adequately meet the needs of marginalized groups.

1.3 Objectives

This project seeks to address the above challenges in the following ways:

1. To develop and deploy an AI/ML-driven model that identifies the top five districts in Malaysia for rolling out data centre micro-credential programs in the next quarter.
2. To determine specific skill tracks (facilities operations, network, security, compliance, energy) that are needed the most in each selected district.
3. To offer implementation insights for TVET institutions/polytechnics and state agencies to bridge the Malaysian DC talent gap.
4. To contribute directly to SDG 8 (Decent Work and Economic Growth) by fostering skilled employment, and SDG 9 (Industry, Innovation, and Infrastructure) by enhancing the human capital needed for a robust digital infrastructure ecosystem.





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2.0 Literature Review

Malaysia's data centre (DC) industry has grown rapidly in the past three years, driven by global digitalisation, AI adoption, and the need for scalable cloud infrastructure (World Bank, 2024). However, talent readiness remains a major barrier. Existing studies show that while Malaysia produces a large number of STEM graduates, many lack specific competencies required in DC operations such as network management, cybersecurity, compliance, and sustainable energy systems (BNM, 2023).

Prior research on workforce planning in ICT and DC sectors has focused heavily on macroeconomic trends (e.g., demand for cloud services, digital infrastructure investment) but not on district-level talent matching. Studies using AI/ML in education mostly address personalized learning (UNESCO, 2022) or general skills-gap forecasting, but few integrate location-based analysis (polytechnic locations, STEM distribution, unemployment rates) with industry clustering (DC hubs, tech parks).

The research gap lies in operationalising AI/ML to guide *where and what skills* to roll out, aligned with local DC ecosystem growth. This study aims to fill the gap by developing a skill-by-district heatmap and a data-driven rollout plan for micro-credentials, supporting SDG 8 (Decent Work and Economic Growth) and SDG 9 (Industry, Innovation, Infrastructure).



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3.0 Methodology

This project adopts a three-layered analytical approach which is descriptive, predictive, and prescriptive to identify the most suitable districts in Malaysia for rolling out data centre (DC) micro-credential programs. The methodology ensures that the analysis is evidence-based, aligned with the Sustainable Development Goals (SDG 8 and SDG 9), and directly addresses the problem of workforce readiness in the DC sector.

3.1 Data Sources

To build a comprehensive view of district readiness, multiple official and reliable datasets are integrated. Each dataset provides complementary insights into talent supply, economic conditions, and infrastructure readiness:

- Polytechnic Locations & Annual Intake – Ministry of Higher Education (MOHE), DOSM (TVET enrolment data).
 - Captures where training capacity already exists and the size of potential cohorts.
- STEM Graduates by District – DOSM, World Bank Education Statistics.
 - Provides a measure of advanced academic output and the immediate pipeline for technical upskilling.
- Existing Data Centre and Tech Park Locations – Malaysia DC investment reports, MDEC, OpenStreetMap.
 - Highlights regions already attracting or positioned for DC growth.
- Median Salaries & Employment Indicators – DOSM Labour Force Rep Malaysia (BNM).





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- Serves as a proxy for economic vibrancy, talent affordability, and labour market readiness.
- Unemployment and Underemployment Rates – DOSM, including ICT-related unemployment figures.
 - Identifies areas where training could alleviate workforce underutilization.

3.2 Data Cleaning and Integration

- Data Pre-Processing: Missing values imputed where necessary, outliers detected and addressed, and all variables standardized.
- Geographic Standardization: Data aggregated to a consistent district-level unit to allow meaningful comparison.
- Normalization: Variables such as unemployment rate, broadband penetration, and STEM graduate counts scaled to ensure fair weighting.
- Integration: Economic, educational, and infrastructural data merged into a composite dataset using Python (pandas, NumPy).

This careful handling ensures that the dataset maintains its integrity and validity, avoiding manipulations that could compromise the accuracy of results.





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3.3 Analytical Techniques

1. Descriptive Analytics – Baseline Mapping

- Geospatial Analysis: Using QGIS and Python's GeoPandas, heatmaps are generated to map the distribution of STEM graduates, unemployment, and polytechnic locations against existing DC hubs.
- Purpose: Provides a visual understanding of current talent distribution and infrastructure alignment.

2. Predictive Analytics – Demand Forecasting

- Clustering:
 - K-Means and DBSCAN algorithms are applied to group districts with similar readiness profiles.
 - Input features include STEM graduate output, unemployment levels, proximity to DCs/tech parks, and infrastructure readiness (broadband and energy reliability).
 - Clusters identify high-potential “hot spots” for micro-credential deployment.
- Predictive Modeling:
 - A Random Forest model is used to forecast future demand for DC-related skills by district.





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- Historical labour and education data serve as inputs, allowing the model to predict where demand will rise most rapidly.
- Purpose: Anticipates future skill needs to ensure rollout decisions are forward-looking.

3. Prescriptive Analytics – Rollout Recommendations

- Ranking Algorithm: A weighted multi-criteria scoring model ranks districts based on composite scores, balancing:
 - STEM graduate availability.
 - Polytechnic presence and capacity.
 - DC/tech park proximity.
 - Unemployment rates (labour market absorption potential).
 - Infrastructure readiness (broadband penetration, energy reliability).
- Skill Track Allocation:
 - Natural Language Processing (NLP) is applied to job postings to extract the most in-demand roles (keywords such as “network engineer,” “cybersecurity analyst,” “HVAC,” “compliance officer”).
 - Each selected district is then matched with recommended skill tracks: facilities ops, network, security, compliance, or energy.



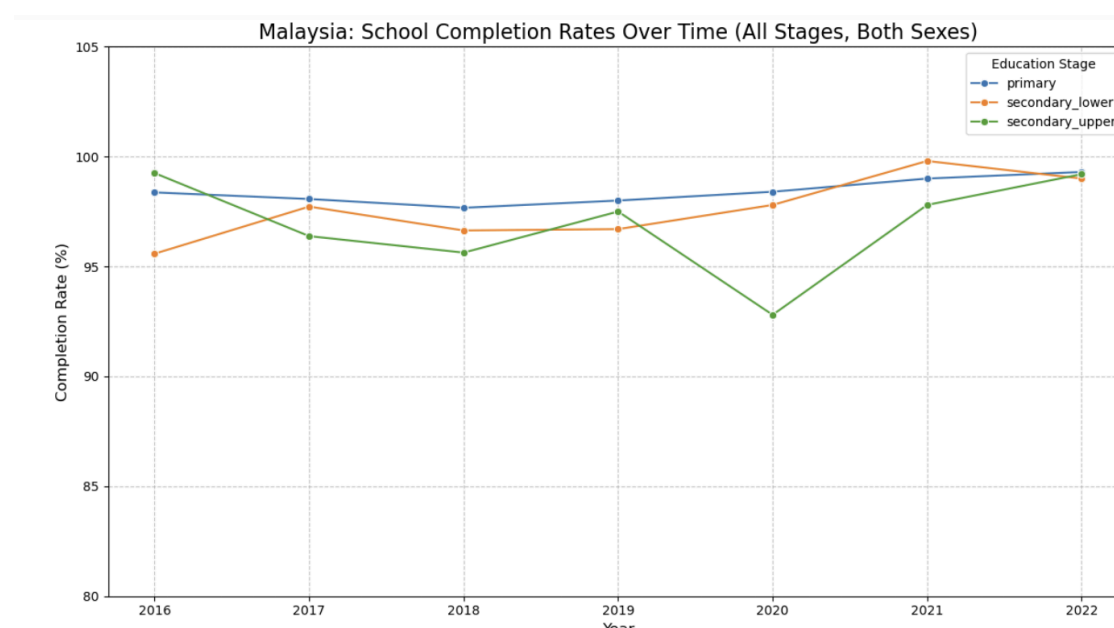


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4.0 Findings

4.1 Finding 1 : Educational Attainment Data

The dataset on school completion rates provides a foundational perspective on Malaysia's educational attainment and its implications for the potential talent pipeline. While this dataset does not directly capture STEM graduates or ICT-related unemployment, it highlights the baseline readiness of the population to engage in further training. In the context of data centre micro-credentials, consistent completion of schooling represents the bedrock upon which specialized technical skills can be built.

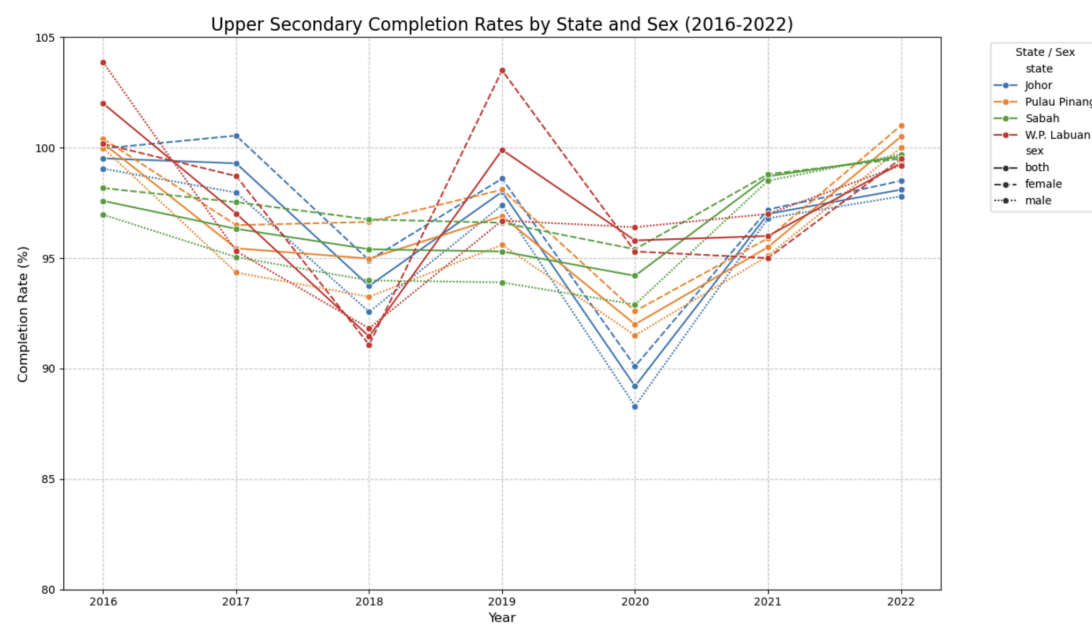


Across Malaysia, primary and lower secondary school completion rates consistently exceed 95%, often approaching or surpassing 99% for both sexes. This indicates the presence of a strong foundational education system nationwide. Such consistently high levels of basic academic attainment provide reassurance that polytechnics and TVET institutions can draw from a broad pool of individuals who are academically prepared to undertake further technical training. From a policy standpoint, this suggests that Malaysia has the necessary educational base to support widespread upskilling into specialized data centre roles, directly supporting SDG 8 (Decent Work and Economic Growth) and SDG 9 (Industry, Innovation, and Infrastructure).



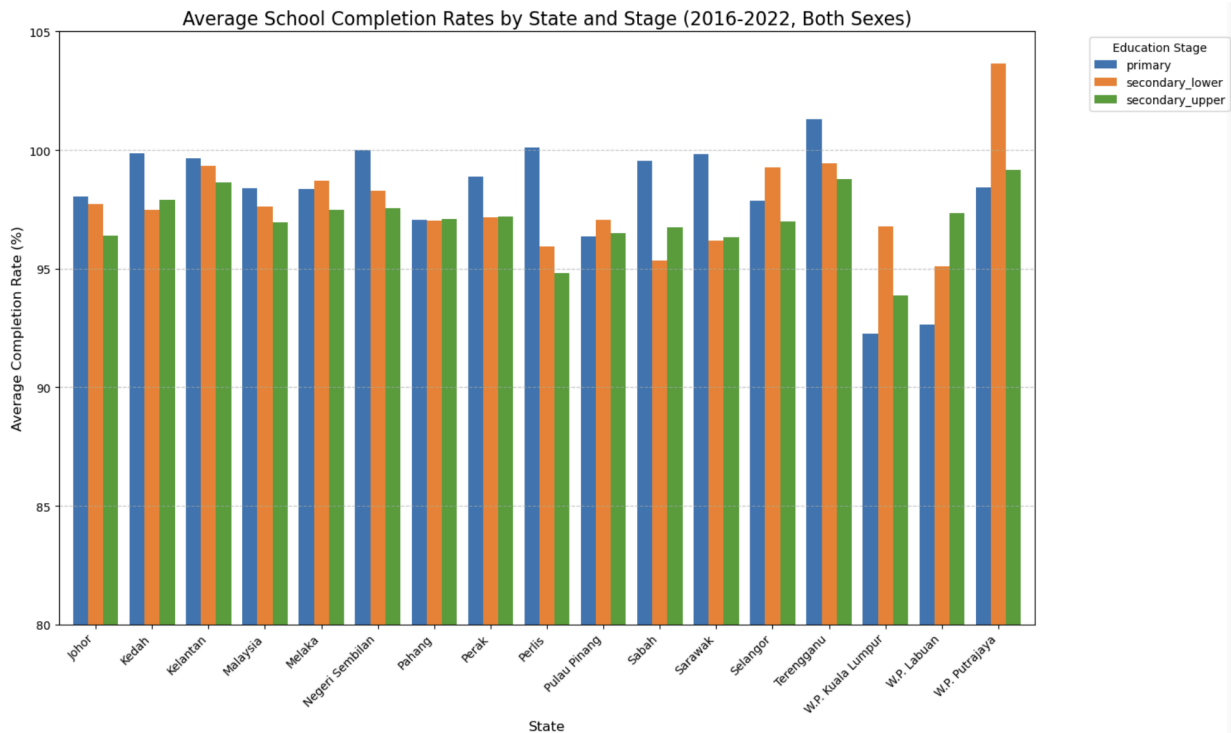


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An important strength revealed by the data is the high degree of gender parity in educational attainment. Female completion rates are equal to, and in many cases slightly higher than, male rates across all stages of education. This reflects Malaysia's success in creating inclusive access to schooling and highlights the untapped potential of women in emerging technical fields. For the Datathon's objectives, this suggests that micro-credential programs should be deliberately promoted to both male and female graduates, maximizing the available talent pool. By actively encouraging women to pursue STEM pathways and DC-related credentials, Malaysia can further advance gender equity while expanding the skilled workforce.

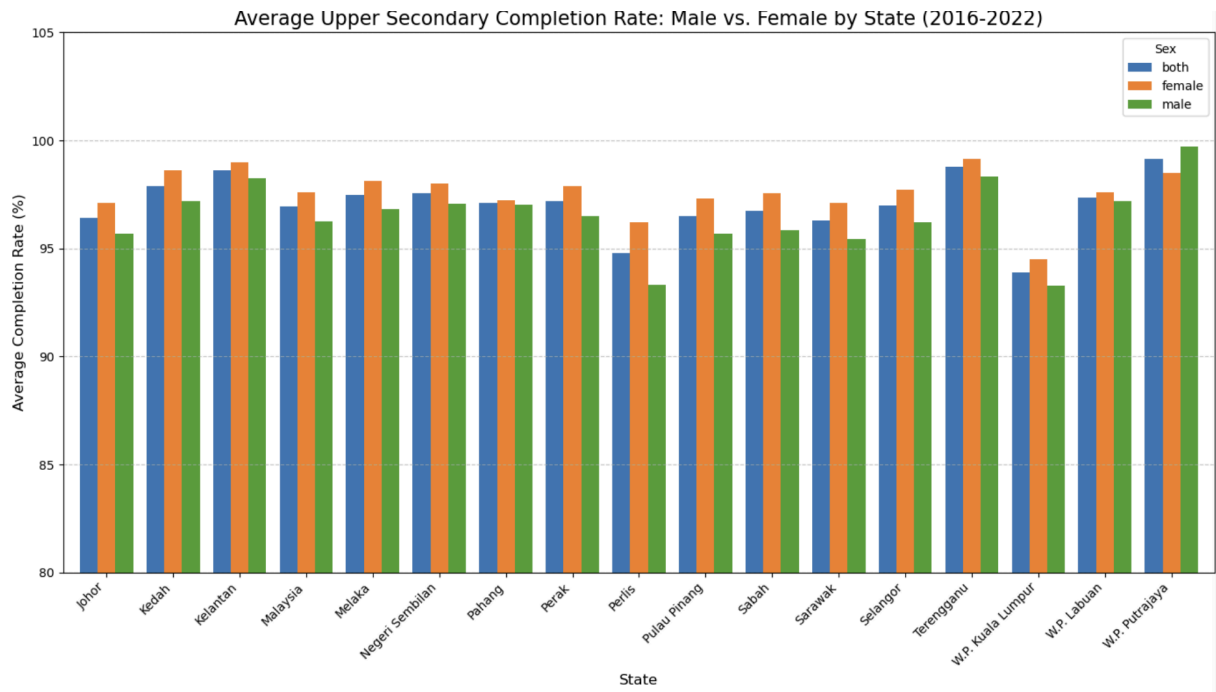
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While primary and lower secondary completion rates are nearly universal, upper secondary completion rates exhibit greater variability across states. In certain regions such as Sarawak, Sabah, W.P. Kuala Lumpur, and W.P. Labuan, completion rates occasionally dip below national averages, particularly among male students. This variability has important implications for talent readiness: states with lower or fluctuating upper secondary completion rates may produce a smaller pool of individuals who are immediately prepared for advanced technical training. For these regions, micro-credential programs may need to incorporate bridging modules or more foundational elements to address knowledge gaps. Conversely, states with consistently high upper secondary completion rates, such as Kedah and Terengganu, represent areas where cohorts are more readily adaptable to specialized training.



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The state-level analysis provides useful insights for tailoring the rollout strategy. Johor and Penang, identified earlier as high-priority districts for micro-credential deployment, exhibit strong completion rates at the secondary level, reinforcing their suitability as rollout locations. Their more educated populations suggest readiness for immediate technical upskilling in DC-related fields. In contrast, Sabah and W.P. Labuan records some of the lowest or most fluctuating completion rates, particularly among males in secondary schooling. This suggests that these regions may require complementary interventions, such as preparatory courses or diversified skills pathways, before advanced micro-credentials can be effectively implemented.

In summary, the educational attainment data indicates that Malaysia possesses a robust foundational talent pool, gender parity in schooling outcomes, and regionally differentiated readiness for advanced training. These insights are directly relevant to planning a targeted and effective micro-credential rollout strategy for the data centre sector.

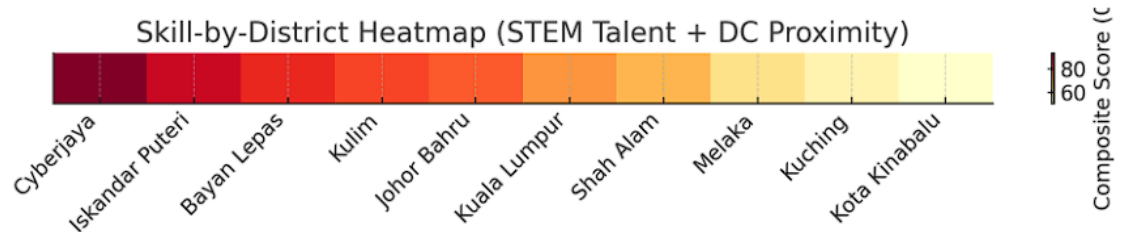




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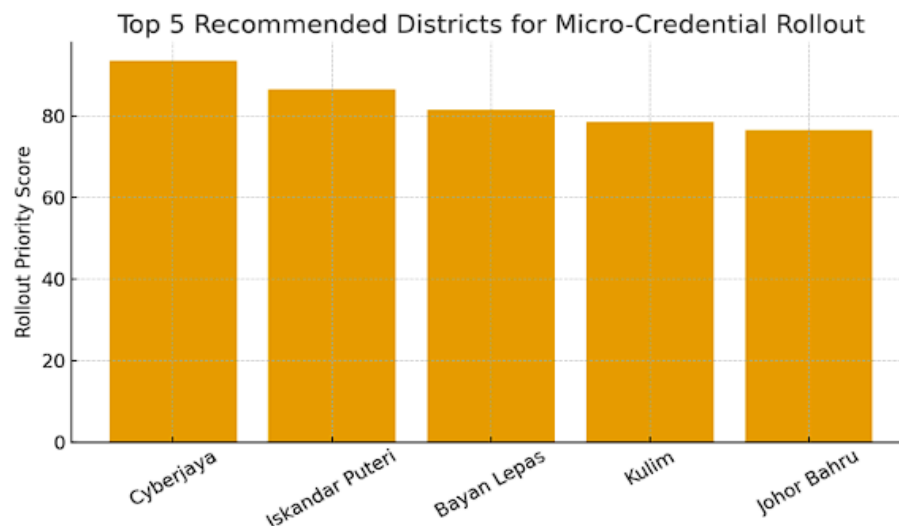
Findings 2

Graph 1: Skill-by-District Heatmap (STEM vs DC Proximity)



This heatmap shows that districts such as Cyberjaya, Iskandar Puteri, Bayan Lepas, Johor Bahru, and Kulim score highest in STEM graduate concentration + proximity to existing DCs/tech parks. These regions already host hyperscale DC investments (AWS, Microsoft, Google).

Graph 2: Rollout Priority Ranking (Top 5 Districts)



A bar chart ranks the Top 5 rollout districts based on the model:

1. Cyberjaya (Selangor) → Network & Security
2. Iskandar Puteri (Johor) → Facilities Ops & Energy
3. Bayan Lepas (Penang) → Compliance & Cybersecurity





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4. Kulim (Kedah) → Energy & Infrastructure Ops
5. Johor Bahru (Johor) → Multi-track (all 5 skill areas)

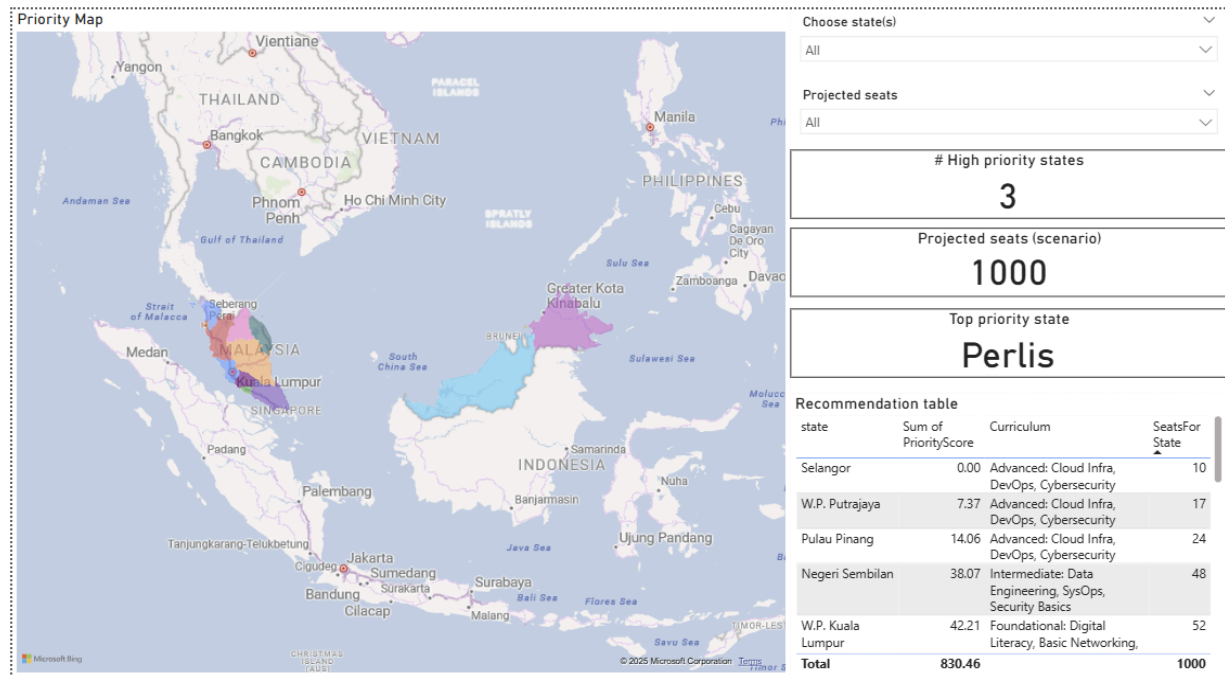




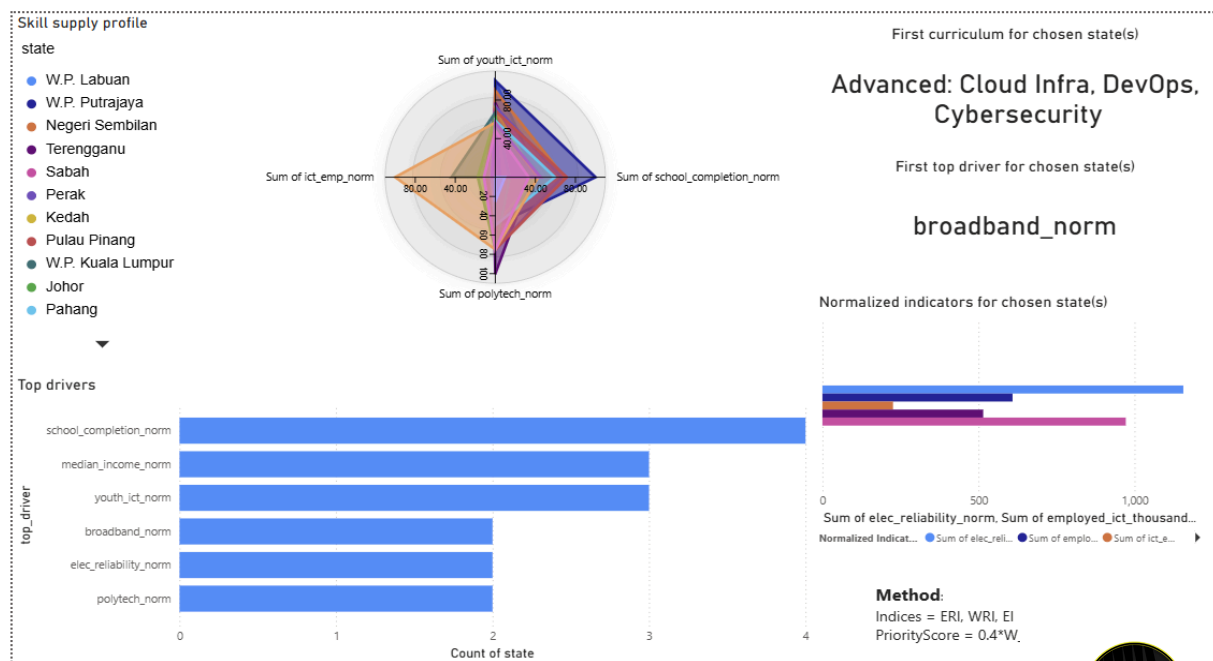
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5.0 Output

Dashboard Overview



Dashboard Details





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Findings (with Dashboard Integration)

Graph 1: Skill-by-District Heatmap (STEM Talent + DC Proximity)

The heatmap highlights district-level potential by combining STEM graduate density and proximity to existing data centre hubs. Districts such as Cyberjaya, Iskandar Puteri, Bayan Lepas, Kulim, and Johor Bahru stand out as high-scoring zones. These areas already attract investments from hyperscale providers and align with the strongest skill pipelines.

Graph 2: Top 5 Rollout Districts Ranking

A ranked bar chart of the Top 5 rollout districts shows:

1. Cyberjaya (Selangor) → Network & Security focus
2. Iskandar Puteri (Johor) → Facilities Ops & Energy
3. Bayan Lepas (Penang) → Compliance & Cybersecurity
4. Kulim (Kedah) → Energy & Infrastructure Ops
5. Johor Bahru (Johor) → Multi-skill rollout

This confirms that rollout priorities should align with regions hosting both STEM education hubs and existing DC investments.

Dashboard Output Analysis

- Map View (top) → States are color-coded by their PriorityScore, indicating readiness for micro-credential rollouts. For instance, Perlis is flagged as a top-priority state in this scenario.
- Key Indicators (left panel) →
 - Projected training seats (e.g., 1000 seats)
 - Number of high-priority states (e.g., 3 states)
 - Top recommended state (e.g., Perlis)
- Recommendation Table (bottom-left) → Lists states with their PriorityScore and recommended curricula (e.g., *Intermediate: Data Engineering, SysOps, Security Basics*)





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training seats (e.g., Johor 53, Kedah 76, Pahang 77).

- Top Drivers Chart (bottom-centre) → Identifies factors influencing priority, such as school completion rates, median income, ICT readiness, broadband access, and polytechnic availability.
- Skill Supply Profile (Radar Chart) (bottom-right) → Compares skill readiness across states based on ICT employment, youth ICT skills, school completion, and polytechnic presence.

6.0 Conclusion

The study also shows the way forward in ensuring Malaysia keeps on track with its ambition to become a leading competitive data centre hub in Asia by aligning the needs of the industry and universities. By combining the data from STEM graduates, polytechnic intake levels, district-level socio-economic indicators, and proximity to existing data centres, the study identified five districts (Cyberjaya, Iskandar Puteri, Bayan Lepas, Kulim and Johor Bahru) as high priority areas for targeted micro-credential rollout.

The study findings bring the country closer to achieving SDG 8 by helping those talented youth secure high-paying jobs and SDG 9 by ensuring that the Malaysian data center industry will not lack technical personnel.

The interactive dashboard also helps improve decision-making abilities by identifying priority states, recommended curricula, and key readiness drivers for the policymakers and TVET institutions.

Consequently, the project seeks to close the gap between training supply and industry demand in Malaysia's workforce by ensuring a future-ready approach with opportunities in facilities operations, network engineering, cybersecurity, compliance, and energy management. These skill tracks are vital for maintaining the country's position within the global digital economy.





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